Tula: Product Doc

Personal health assistant. Version 1.0, 8 Oct 2026. Owner: Alok Kumar Mishra.

Summary

Tula is a personal health assistant that turns a meal photo or a blood report into a clear daily plan. The user spends about 10 seconds a day giving input; Tula calculates, explains and nudges. The version built in this assignment covers four flows: onboarding that ends in a personal plan, effortless meal logging, lab reports that change daily targets, and a proactive Today screen with chat. Login, fitness integrations and social features are deliberately cut.

1. Problem

People who want to eat better or fix a lab value usually fail for three reasons:

1. **Logging is too much work.** Calorie apps need searching a database and weighing portions. Most users stop within a week.
2. **Indian food is badly served.** Global food databases handle "2 rotis, dal, aloo sabzi and a katori of curd" poorly, so estimates are wrong and users lose trust.
3. **Data has no meaning.** A blood report sits in a WhatsApp chat. A calorie number has no link to why it matters for this person. Without a reason, motivation fades.

Existing apps store data. They don't tell the user what to do next or why.

2. Target user

Primary persona: Rohan, 31, software engineer in Bengaluru. Gained 8 kg since joining his job. His last blood test showed LDL at 152 mg/dL and borderline HbA1c. He has tried HealthifyMe and MyFitnessPal and quit both in under two weeks. He eats home food, office canteen lunches and Swiggy dinners. He will give the app a few seconds per meal and nothing more.

Secondary persona: Priya, 38, manager with a thyroid condition and low Vitamin D. Wants to understand her reports and make small diet changes. Less interested in calorie counting, more in "what should I change".

Jobs to be done:

- When I eat, I want to record it without effort, so I know where I stand for the day.
- When I get a blood report, I want to understand it in plain words and know what to change.
- During the day, I want to be told what to do next, so I don't have to plan it myself.

3. Product principles

1. **10 seconds a day.** If a feature adds daily effort, it needs a strong reason to exist.
2. **Every number has a why.** Targets are explained with the user's own data.
3. **The assistant reaches out first.** Reminders and nudges come from Tula; the user doesn't have to remember to open the app.

4. **Honest estimates.** Show confidence, ask one question when unsure, and let the user correct anything in one tap.
5. **Safe by design.** No diagnosis, a calorie floor, a cap on weekly loss rate and a doctor referral line on flagged values.

4. Scope and prioritisation

4.1 Flows built

Priority	Flow	Why it is core
P0	Onboarding to personal plan	Answers "what are my goals and why". Without it, the rest is a logging tool.
P0	Meal logging by photo, voice or text	The daily habit. If this is slow, users leave in a week.
P0	Today screen with proactive nudge	Turns logs into "what next". This is where the assistant feels like one.
P1	Lab reports linked to goals	The differentiator. Gives a personal WHY and changes targets.
P1	Chat with Tula, grounded in user data	Covers questions the screens don't. Also a second way to log.
P2	Local reminders and evening recap	Low cost, large effect on day-two and day-seven retention.

4.2 Deliberately cut

Cut	Reason	What replaces it
Login and accounts	Not core to the health problem; brief allows skipping	Anonymous device ID stored securely, plus seeded demo profiles
Fitness and wearable sync	Large scope, many integrations, weak link to the core loop	Activity level set once in onboarding
Full food database and barcode scan	Indian home food rarely has barcodes; a database would be weeks of work	Vision model estimates plus a small curated table of common Indian dishes
Separate profile screen	Progress already lives on Today	Plan editable from the Today header
Social, challenges, leaderboards	Health is personal; adds moderation and scope	Streaks and private milestones
Server push notifications	Needs push infrastructure for little gain in a demo	On-device scheduled notifications
Water and sleep tracking	Adds daily effort with little insight at this stage	Water target shown as a static tip

5. Feature specification

5.1 Onboarding and personal plan

Inputs: goal, name, age, sex, height, weight, activity level, optional target weight, optional lab report.
Calculation (deterministic, not by the language model):

- BMR by Mifflin-St Jeor; TDEE = BMR multiplied by an activity factor (1.2, 1.375, 1.55, 1.725).
- Lose weight: TDEE minus 500 kcal, never below 1,200 kcal (women) or 1,500 kcal (men), and weekly loss capped at 0.75 kg.
- Build muscle: TDEE plus 250 kcal; protein 1.6 g per kg body weight.
- Default protein 1.2 g per kg; fibre 25 to 30 g; water 35 ml per kg.
- Lab adjustments: high LDL raises fibre to 30 g and caps saturated fat; high HbA1c caps added sugar and shifts carbs earlier in the day; low Vitamin D and B12 add food suggestions.

Output: daily calorie and macro targets, projected date to reach target weight, and two or three sentences of personal reasoning written by the language model from the calculated numbers.
Acceptance criteria: completes in under 60 seconds; resumes after the app is killed; unrealistic goals (BMI under 18.5 after loss, or more than 1 kg a week) are adjusted with an explanation.

5.2 Meal logging

Inputs: photo (camera or gallery), voice (up to 30 seconds, English, Hindi or Hinglish) or typed text.
Behaviour:

1. Meal type is guessed from the time of day.
2. Input is parsed into items with name, portion, unit, kcal, protein, carbs, fat, fibre and a confidence score.
3. Common Indian dishes are matched to a curated nutrition table so that "1 katori dal" always gives a consistent value.
4. The user reviews and edits on the confirmation sheet, then logs.
5. The Today ring updates and a one-line insight appears.

Edge cases:

- Not food: polite message, retake or type.
- Low confidence (under 0.6): one clarifying question with two or three options.
- AI timeout or rate limit: save the raw input as a pending meal and enrich it in the background.
- Offline: queue locally with the photo; sync when online.
- Duplicate submit: idempotency key, so a double tap logs once.
- Late logging: allow choosing the date and time for a meal from earlier today or yesterday.
- Edit and delete: any meal, with Undo for five seconds after delete.

Shortcut: "Same as yesterday" and the two most frequent meals are offered as one-tap chips.

5.3 Lab reports

Inputs: PDF or photo of a blood report, up to 10 MB.
Supported markers in this version: HbA1c, fasting glucose, total cholesterol, LDL, HDL, triglycerides, Vitamin D, Vitamin B12, haemoglobin, TSH, creatinine, uric acid.
Behaviour: extract values and units, convert units to a standard (for example mmol/L to mg/dL), compare against the report's range or a standard range, and flag each marker as low, in range or high. Flagged markers get a plain explanation, a personal reason and one daily action that changes a target.
Edge cases: not a lab report, unreadable scan, partial extraction (show what was found and let the user add a missing value by hand), unknown marker (store but don't explain), duplicate upload (detect by file hash), and values that look like extraction errors (for example HbA1c of 65) held for user confirmation.
Safety: no diagnosis. Critical values (for example fasting glucose above 250 mg/dL) show a clear "Please see a doctor soon" message instead of diet tips.

5.4 Today and proactive nudges

The nudge engine picks at most one nudge at a time using rules, then the language model writes the wording.

Rule	When	Example
Missing meal	Meal window passed with nothing logged	Lunch not logged yet. Snap it now?
Protein gap	After 4 pm and protein under 50 percent	40 g protein to go. Curd or roasted chana would help.
Fibre gap (LDL users)	After 6 pm and fibre under 60 percent	Add a salad or sprouts at dinner for your fibre target.
Calorie room	Before dinner	620 kcal left. A normal thali fits.
Streak protection	8 pm and today not on track	Log dinner to keep your 5-day streak.
Day complete	All meals logged and on target	Day complete. Same again tomorrow.

Local notifications: meal reminders only if that meal isn't logged, an evening recap at 9 pm, and a Sunday weekly summary. Maximum three a day. User can turn each off.

5.5 Chat with Tula

Chat answers questions using the user's own data through server-side tools: get today's summary, get the week's history, get report markers, and log a meal. It never invents numbers it didn't get from a tool. It declines diagnosis and medication questions and points to a doctor.

5.6 Motivation

- Streak counts days with at least two meals logged; one free rest day a week.
- Milestones at 3, 7, 14 and 30 days.
- Weekly recap: days on target, average protein, best day, one thing to try next week.
- Marker progress: when a new report shows an improved value, a dedicated celebration card.

6. Key user journeys

Day 1

1. Install, onboard in under a minute, optionally upload a report.
2. See the plan with a personal reason.
3. Log the first meal by photo; the ring moves.
4. Get a lunch reminder at 1:30 pm because lunch isn't logged.
5. Evening recap: "Day 1 done. Protein was low; tomorrow try eggs or paneer at breakfast."

Day 7

1. Morning: open the app from a reminder, tap "Same breakfast as yesterday".
2. Streak hits 7; small milestone.
3. Sunday recap shows 5 of 7 days on target and one suggestion for next week.

7. Success metrics

Metric	Target for a real launch	Why
Time to log a meal (median)	Under 10 seconds	The core effort promise
Onboarding completion	Over 80 percent	Plan is the first value moment
Meals logged per active day	2.5 or more	Enough data for useful insights
Day-7 retention	Over 35 percent	Category average is far lower
Edit rate on AI meal items	Under 25 percent	Proxy for estimate accuracy
Nudge action rate	Over 20 percent	Nudges are useful, not noise

For the assignment, log these as local analytics events so the instrumentation is visible in code.

8. Self-assessment against the brief

Question from the brief	How Tula answers it
Am I onboarded well, with clear goals?	Plan screen with targets, projection and reasons
Clear WHAT and WHY?	Targets explained with the user's own numbers and lab values
Is the assistant proactive?	Nudges, reminders only when needed, repeat-meal shortcuts, chat logging
Motivation and feel-good signals?	Ring, streak with rest day, milestones, weekly recap, marker improvement card
Intuitive and not confusing?	Three tabs, one add button, one nudge at a time, short copy
Would I open it tomorrow?	Evening recap sets up tomorrow; morning nudge gives the first action

9. Risks and open questions

- **Estimate accuracy for mixed Indian plates.** Mitigation: curated dish table, confidence scores and one-tap portion edits.
- **Free model rate limits during the demo.** Mitigation: provider fallback, caching by image hash, and a pending-meal state.
- **Health data privacy.** The free model tier may use inputs for training; acceptable for an assignment, not for production. Documented in the README.
- **Notification fatigue.** Hard cap of three a day and per-type toggles.
- Open: should "Improve a health marker" users see calories at all, or only macros and fibre? Default for now: calories shown smaller.

10. Future roadmap (not in this build)

- Wearable step and sleep sync
- Restaurant and Swiggy order import
- Family profiles
- Doctor-shareable monthly summary PDF
- Regional language voice input